

When mutualism turns costly: abiotic stress shifts endophyte benefits in cool-season grasses

Jacob K. Moutouama ^{*1,2,3}, Julia Martin¹,
Ulisses Rizo¹, Malcolm Sherwood¹, Emily Chong¹, Dajanae Pearson¹, Alexandra Jimenez
Martín¹, Josh Fowler^{1,4}, Ali Campbell¹, Chris Oxley¹, Karl Schrader¹, and Tom E.X. Miller ¹

¹Program

in Ecology and Evolutionary Biology, Department of BioSciences, Rice University, Houston, TX USA

²Department of Botany, University of British Columbia, Vancouver, BC, Canada

³University of Miami, Department of Biology, Miami, FL USA

Running header: Stress and Grass–Endophyte Interactions

Keywords: demography, mutualism, parasitism, plant–fungi interactions, range limits, stress gradient hypothesis

Submitted to: *Ecology letters* (Letter)

Data accessibility statement: If the paper is accepted, all data and computer scripts supporting the results will be archived in a Zenodo package, with the DOI included at the end of the article. During peer review, our code (Stan and R) is available at <https://github.com/jmoutouama/endo-range-limits>.

Conflict of interest statement: None.

Authorship statement:

Abstract:

Main Text:

Figures: 6

Tables: X

References: X

*Corresponding author: jmoutouama@gmail.com

Abstract

Beneficial interactions between plants and fungi are ubiquitous and foundational to terrestrial ecosystems. Plant-fungal mutualisms are expected to strengthen under environmental stress, yet the relative importance of abiotic and biotic factors in eliciting mutualistic benefits remains unclear. We studied how the demographic effects of seed-transmitted fungal endophytes on native three grass host species varied in response to abiotic and biotic factors using geographically distributed field experiments along an aridity gradient, cross with herbivore access or exclusion. Vital rate models showed that all components of host fitness (survival, growth and fertility) declined with increasing, indicating reduced performance under abiotic stress. Declines were more pronounced in individuals hosting endophytes, suggesting that interactions that are beneficial under benign conditions can become costly under abiotic stress. On the other hand, herbivory had no detectable effect on host demography or on the demographic effects of hosting endophytes. Our results indicate that abiotic stress was the primary determinant of interaction outcomes between plants and fungal symbionts, but in the opposite direction as predicted by theory and prior work. These results highlight that symbiotic fungi can become liabilities under stress, with implications for the dynamics of plant-fungal mutualisms under global change.

17 Introduction

18 Plant-microbe symbioses are widespread and ecologically important. These interactions are
19 famously context-dependent, where the direction and strength of the interaction outcome
20 depend on the environment in which it occurs (Bronstein, 1994; Hoeksema and Bruna, 2015).
21 For example, plant-microbe interactions that are beneficial under stressful conditions may
22 become neutral or parasitic under benign conditions (Giauque et al., 2019). Variation in the
23 nature of biotic and abiotic stress is a key driver of these context-dependent outcomes.

24 Microbial symbioses can substantially enhance plant resilience to environmental stress.
25 Under biotic stress, such as herbivory, microbial symbiosis can benefit host plants by
26 facilitating the production of secondary compounds that deter feeding or exert direct toxicity,
27 thereby reducing herbivore growth, survival, and oviposition (Atala et al., 2022; Bastias et al.,
28 2017; Vega, 2008). Similarly, under abiotic stress, such as low precipitation, nutrient limitation,
29 or high temperature, symbiotic microbes can enhance host tolerance through a variety of
30 mechanisms (Afkhami and Rudgers, 2008; Clay and Schardl, 2002). Symbionts including myc-
31 orrhizal fungi, rhizobia, and endophytic bacteria can improve water and nutrient acquisition by
32 increasing root surface area, facilitating nutrient mobilization, or promoting nitrogen fixation,
33 allowing plants to maintain growth and metabolic function under drought or poor soil
34 conditions (Amijee et al., 1989). In addition, many symbiotic microbes stimulate the production
35 of antioxidant enzymes, such as superoxide dismutase and catalase, which reduce oxidative
36 damage caused by stressors like drought, salinity, and heat (Ruiz-lazona et al., 1996). By
37 improving resource acquisition, modulating plant physiology, and enhancing stress-responsive
38 metabolism, these symbiotic interactions help host plants buffer against environmental
39 stochasticity and extreme conditions across a wide range of ecosystems (Fowler et al., 2023).

40 Although microbial symbioses often enhance plant performance, they can also impose
41 substantial costs, particularly in resource-limited environments. Maintaining symbionts
42 requires hosts to allocate valuable carbon and nutrients to sustain their microbial partners.
43 For instance, mycorrhizal fungi may require up to 20% of photosynthetically fixed carbon
44 (Soudzilovskaia et al., 2015; Thirkell et al., 2020), and under abiotic stress, this investment
45 can outweigh the benefits of the symbiosis (Bronstein, 2001; Johnson et al., 1997), reducing
46 host survival, growth, reproduction, or competitive ability (Klironomos, 2003). Similarly,
47 vertically transmitted endophytes can shift from mutualistic to antagonistic interactions when
48 they produce stromata that abort host inflorescences or impose energetic costs (Saikkonen
49 et al., 2004). Experimental studies further show that under low nutrient or water conditions,
50 endophyte-infected grasses may produce fewer tillers and lower total biomass compared to
51 endophyte-free plants, reflecting the costs of sharing limited resources (Ahlholm et al., 2002).

52 Despite growing interest in the role of symbiosis in shaping host performance, few
53 studies have tested under natural conditions how abiotic and biotic stressors synergistically
54 influence the demographic effects of symbionts along environmental gradients. Most existing
55 work has been conducted in controlled or common garden experiments, where, for example,
56 herbivory but not drought reduced plant growth, and no interaction between the two stressors
57 was observed (Emery et al., 2010). Studying the combined effects of abiotic and biotic stressors
58 under natural conditions is particularly important in the context of global change (Louthan
59 et al., 2015), as shifting climate patterns and altered herbivore pressures may change the
60 balance of costs and benefits in plant–microbe interactions, potentially influencing species
61 distributions and ecosystem function. Understanding how abiotic stress, biotic stress, and
62 endophyte infection interact requires experiments that jointly vary all three factors. However,
63 to our knowledge, no such comprehensive study has been conducted in native grass species.
64 Here, we address this gap by examining the additive effects of biotic and abiotic stress on
65 grass–endophyte symbioses in native populations.

66 Grass endophytes, especially those vertically transmitted through seeds, provide a
67 tractable system for studying symbiosis, as infection status (symbiont-free vs. symbiont-
68 infected) can be experimentally manipulated. A well-studied example is the association
69 between cool-season grasses and *Epichloë* fungi (Clay et al., 2005; Oberhofer et al., 2014; Saikkonen
70 et al., 2010; Schardl et al., 2004). Endophyte-infected grasses often perform better than
71 endophyte free grasses under water limitation, although benefits can vary with host, endophyte
72 species, or genotype (Saikkonen et al., 1998). Beyond drought tolerance, infection frequency it-
73 self is influenced by herbivory and environmental conditions, including long-term variation in
74 precipitation and temperature (Clay et al., 2005; Fowler et al., 2025; Rudgers et al., 2016). Even
75 when endophytes do not enhance survival under stress, they may still boost host fitness by pro-
76 moting growth and reproduction (Yule et al., 2013). Together, these features make grass–*Epichloë*
77 symbioses an ideal system for investigating how abiotic stressors, such as drought, interact with
78 biotic stressors, like herbivory, to determine the outcome of symbiosis in cool-season grasses.

79 Across an aridity gradient in the south-central United States, we investigated how
80 endophyte symbiosis influences host-plant demography under varying abiotic and biotic
81 stress. This region spans a pronounced east–west precipitation gradient, from mesic conditions
82 in the east to increasingly arid conditions in the west, providing a biologically realistic stress
83 gradient. At each site, we manipulated herbivory to assess its potential interaction with
84 abiotic stress. By experimentally removing the endophyte in some individuals, we examined
85 the symbiosis between three native cool-season grasses (*Agrostis hyemalis*, *Elymus virginicus*,
86 and *Poa autumnalis*) and their vertically transmitted fungal symbionts (*Epichloë* spp.). We
87 addressed two questions: (i) under combined abiotic and biotic stress, do endophytes provide

fitness benefits or impose costs on their host plants? (ii) which factors (abiotic vs. biotic) determine whether the grass–endophyte interaction benefits or harms the host? We predicted that endophytes would enhance host performance under low precipitation by mitigating drought stress (Fig. 1A; stress-amelioration scenario). Specifically, in fenced plots with reduced herbivory, endophyte benefits are expected to be smaller because the contribution of symbionts to host performance is less critical when biotic stress is low. In contrast, under high abiotic stress combined with herbivore presence, the grass–endophyte symbiosis may become less mutualistic, as herbivory can suppress or interfere with the positive effects of endophytes and could even impose a net cost (Fig. 1B; inhibition scenario). Furthermore, while both abiotic and biotic stress are expected to influence the outcome of endophyte–grass symbiosis, we anticipate that abiotic stress will be the primary driver of mutualistic benefits, with biotic stress modulating the magnitude or direction of those benefits.

Materials and methods

Study system

Our study focused on three cool-season perennial grasses native to woodland and prairie habitats of eastern North America: *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* (subfamily Pooideae) (Shaw, 2011). *A. hyemalis* is a small, short-lived perennial that germinates from autumn to late winter and typically flowers between March and July. *E. virginicus* is a larger, long-lived species that begins flowering as early as March or April and continues throughout the summer. *P. autumnalis* has a similar growth form to *E. virginicus* and flowers from late spring through summer. All three species host *Epichloë* fungal endophytes, which are primarily transmitted vertically through seeds. *Agrostis hyemalis* is associated with *Epichloë amarillans* (White Jr, 1994), *E. virginicus* typically hosts *Epichloë elymi* (Liu, 2023), and *P. autumnalis* can also harbor endophyte taxa capable of horizontal transmission via sexual stromata on inflorescences or asexual conidia on leaf surfaces (Clay and Schardl, 2002; Gundel et al., 2020; Leuchtman et al., 2014). However, horizontal transmission is rare, influenced by environmental and genetic factors, and was not observed in our study system.

To examine the effects of endophytes on host species across a stress gradient, we established eight plots (1.5 m × 1.5 m) at each of seven common garden along the aridity gradient from Louisiana and Texas (Fig. 2A–D). Each plot contained 15 individuals of a single species, with half of the plots protected from herbivores using exclusion fences to assess the interactive effects of endophytes and herbivory (Fig. 2E). In summer–fall 2021, seeds were collected from naturally symbiotic source populations and either heat-treated to remove

endophytes or left untreated, producing symbiont-free (E^-) and symbiotic (E^+) plants from the same genetic lineages.

Source populations and symbiont removal

Plants used in the common garden experiment were derived from natural (source) populations throughout the natural distribution of the species (Fig. 2, Table S1). Seeds were collected from these source populations, and some were heat-treated at 60°C for approximately five days (120 hours) to produce endophyte-negative plants (E^-), a method that eliminates endophytes without affecting seed viability. All seeds, both heat-treated and non-heat-treated, were planted in the Rice University greenhouse, where seedlings were fertilized every two weeks. Seedlings were then vegetatively propagated to produce enough individuals for the experiment. Before field planting, we confirmed endophyte status (E^+ or E^-) of all seedlings using either microscopy or an immunoblot assay. Leaf tissues were stained with aniline blue lactic acid and viewed under a compound microscope at 200x–400x to identify fungal hyphae. The immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co.) uses monoclonal antibodies that target proteins of *Epichloë* spp. and a chromagen to visually indicate presence or absence (Koh et al., 2006). Both methods yield similar detection rates.

Common garden experimental design and climate data

We conducted our common experiment across spanning the precipitation gradient from Louisiana to central Texas, USA (Fig. 2, Table S-1). To characterize the abiotic stress gradient, we collected daily temperature and precipitation data from PRISM (Hart and Bell, 2015). These daily values were used to calculate the mean temperature (°C) and cumulative precipitation (mm) for each year, spanning from the time plants were transplanted to when demographic data were collected. During our study period (2023–2025) realized precipitation at these sites corresponded well to 30-year normals, with wetter eastern sites and drier western sites. Mean annual temperature did not differ significantly across sites (Fig. S1), so we focus on precipitation as the primary abiotic factor that varied across sites. This gradient also overlaps and exceeds the natural distributions of our focal species, ensuring realistic exposure to climatic and biotic conditions (Fig. 2A,B, C).

In each common garden, we installed eight plots per species. Plots were randomly assigned one of four initial endophyte frequency treatments (80%, 60%, 40%, or 20%) and one of two herbivore access treatments (fenced to exclude herbivores or unfenced to allow access). In the fenced plots, insecticides were applied periodically to eliminate small herbivorous insects, as *Agrostis hyemalis*, *Epichloë amarillans*, and *Poa autumnalis* are naturally subject to

herbivory by both insects and larger mammals. To ensure that our herbivory treatment reflected the level of herbivory experienced by individual plants, we counted the number of damaged plants per plot and estimated the proportion of damaged plants per site and herbivore treatment. Our results confirmed that the herbivory treatment was effective (Fig. 2F).

Demographic data collection

We collected demographic data, including survival, growth, and reproduction, during the flowering season of each species: *Agrostis hyemalis* and *Poa autumnalis* were censused in May, and *Elymus virginicus* was censused in June of 2023, 2024, and 2025. For each individual, plant survival was recorded as alive or dead and, for surviving plants, size was measured as the number of tillers. Growth was quantified as the log ratio of the number of tillers between consecutive censuses (i.e., $\log(\text{tiller}_{t+1}/\text{tiller}_t)$) for plants that were alive in both years. We recorded the number of inflorescences per plant for all three species and counted the number of spikelets on up to three inflorescences from reproducing individuals of *Poa autumnalis* and *Elymus virginicus*.

Assessing whether endophytes confer benefits or impose costs under abiotic and biotic stress

To assess whether endophytes confer benefits or impose costs under abiotic and biotic stress, we first developed two candidate models for each vital rate: one quadratic and one monotonic. All models shared the same hierarchical structure and were applied to survival, growth, and fertility. Survival was modeled with a Bernoulli distribution, growth with a Gaussian distribution, the number of inflorescences with a zero-inflated negative binomial distribution, and the number of spikelets with a negative binomial distribution. We modeled all vital rates (survival, growth, and fertility) using species-specific fixed effects for symbiosis (endophyte presence), precipitation, and herbivore access, including all relevant two- and three-way interactions. Each observation i corresponds to a plant in a given year; individuals can appear in multiple years. For simplicity, we omit the full subscript notation for species, plot, population, and site-year, which is explicitly included in the Stan model. We present the quadratic model here (Eq. 1) because it encompasses the linear model as a special case.

$$\begin{aligned} \mu_i = & \beta_0[Sp_i] + \beta_P[Sp_i]P_i + \beta_{P^2}[Sp_i]P_i^2 + \beta_{endo}[Sp_i]endo_i + \beta_{herb}[Sp_i]herb_i \\ & + \beta_{endo \times P}[Sp_i]endo_iP_i + \beta_{herb \times P}[Sp_i]herb_iP_i \\ & + \beta_{herb \times endo}[Sp_i]herb_iendo_i + \beta_{endo \times herb \times P}[Sp_i]endo_iherb_iP_i \\ & + \phi_{site[i],Sp_i} + \omega_{source[i],Sp_i} + \rho_{plot[i],Sp_i}. \end{aligned} \tag{1}$$

Here, P_i is total precipitation (mm) cumulated over the 12 months preceding the census, $endo_i$ indicates endophyte presence, and $herb_i$ indicates herbivore access. Random effects account for background variation at multiple hierarchical levels: $\phi_{\text{site}[i], Sp_i}$ captures site-to-site heterogeneity, $\omega_{\text{source}[i], Sp_i}$ captures variation associated with the provenance of transplants, and $\rho_{\text{plot}[i], Sp_i}$ captures plot-to-plot variation. Each species has its own random effect for each site, source, and plot, while the variance parameters (σ_{site} , σ_{source} , σ_{plot}) are shared across species. This hierarchical structure allows the model to "borrow strength" across species, improving estimates of fixed effects while still capturing species-specific responses to environmental gradient. We confirmed that models effectively captured key aspects of the grass demography data, including means, standard deviations, and quantiles using a posterior predictive check (Fig. S2).

Secondly, we compared quadratic and monotonic candidate models using leave-one-out cross-validation (LOOCV) to identify the best-supported model for each vital rate (Vehtari et al., 2017). LOOCV combines validation and training by iteratively holding out one observation as a test set while fitting the model on the remaining $n - 1$ observations. This process is repeated n times, once for each observation, resulting in n fitted models (Silva and Zanella, 2024). The overall test error estimate is then obtained by averaging the prediction errors across all n models (Eq. 2) for each vital rate.

$$CV_n = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

Although we evaluated quadratic models, LOOCV showed that they provided minimal improvement over monotonic models (Table S-2). Therefore, all subsequent analyses used the monotonic model, which captures the observed patterns while maintaining model simplicity.

Finally, for each vital rate, precipitation was summarized at key quantiles (10th, 25th, 50th, 75th, and 90th percentiles) to represent the range of site-level abiotic stress. For each species (*Agrostis hyemalis*, *Elymus virginicus*, *Poa autumnalis*) and herbivory treatment (Fenced or Unfenced), we computed posterior predictions of survival probability for individuals with endophytes (E^+) and without endophytes (E^-) using the best Bayesian model outputs (Eq. 1). The difference ($\Delta = E^+ - E^-$) was calculated for each posterior sample to estimate median effects and 90% credible intervals. We also computed the posterior probability that $\Delta > 0$, representing the likelihood that endophytes confer benefits on vital rates under the given climate and herbivory conditions.

Drivers of endophyte effects on cool-season grass demography

To identify the main drivers of endophyte-mediated effects, we fitted three hierarchical Bayesian candidate models for each vital rate (survival, growth, inflorescence, and spikelet production; Supplementary Methods S.1.2). Each model included endophyte status as a baseline predictor and differed in whether it incorporated interactions with precipitation (Endophyte $\times$ Climate), herbivory (Endophyte $\times$ Herbivory), or both factors, including the three-way interaction (Endophyte $\times$ Climate $\times$ Herbivory). Random effects for site-year ($\phi_{\text{site}[i], \text{Sp}_i}$), source population ($\omega_{\text{source}[i], \text{Sp}_i}$), and plot ($\rho_{\text{plot}[i], \text{Sp}_i}$) captured hierarchical variation while accommodating species-specific responses. We compared candidate models using leave-one-out cross-validation (LOO-CV) (Vehtari et al., 2017), which estimates predictive accuracy by iteratively holding out each observation. Differences in expected log predictive density (ΔELPD), along with associated standard errors, were used to rank model performance.

All models were implemented in R (R Core Team, 2023) using Stan via the rstan interface (Stan Development Team, 2024).

Results

Endophyte effects on host fitness components across abiotic and biotic stress gradients

Across fitness components, *Epichloë* endophytes generally enhanced host performance under wetter conditions, whereas their effects were costly or occasionally neutral under drier conditions, consistent with the inhibition scenario (Fig. 1). Benefits were also less pronounced when herbivores had access.

Grass–*Epichloë* symbiosis had mostly negative or neutral effects on survival at low precipitation, but increasingly enhanced survival under wetter conditions (Fig. 3; Fig. S-1). In *Agrostis hyemalis* and *Elymus virginicus*, median $\Delta(E^+ - E^-)$ ranged from approximately -0.026 to 0.019 under low precipitation in unfenced plots ($\text{Pr}(\Delta > 0) = 0.37\text{--}0.61$) and increased to $0.022\text{--}0.106$ under high precipitation ($\text{Pr}(\Delta > 0) = 0.70\text{--}0.85$; Table S-3). In fenced plots, positive effects appeared only at the highest precipitation for *Agrostis* (median $\Delta = 0.011\text{--}0.038$, $\text{Pr}(\Delta > 0) = 0.59\text{--}0.72$), while for *Elymus*, strong positive effects were evident at high precipitation (median $\Delta = 0.089\text{--}0.113$, $\text{Pr}(\Delta > 0) = 0.91\text{--}0.95$). *Poa autumnalis* showed little survival benefit in unfenced plots across the gradient (median $\Delta = -0.056$ to -0.002 , $\text{Pr}(\Delta > 0) = 0.14\text{--}0.43$), but weakly positive effects emerged when herbivores were excluded (median $\Delta = 0.002\text{--}0.026$, $\text{Pr}(\Delta > 0) = 0.57\text{--}0.71$).

Epichloë endophyte presence generally enhanced relative growth of individuals across the precipitation gradient, with benefits increasing under wetter conditions and modulated by herbivory (Table S-4). In *A. hyemalis*, effects were strongly positive even at moderate precipitation in unfenced plots (median $\Delta = 0.370$ – 0.385 ; $\Pr(\Delta > 0) = 0.946$ – 0.992), and similar trends were observed in fenced plots though slightly weaker (median $\Delta = 0.080$ – 0.262 ; $\Pr(\Delta > 0) = 0.661$ – 0.878). *E. virginicus* showed small positive or neutral effects under dry conditions (median $\Delta = 0.232$ – 0.259 ; $\Pr(\Delta > 0) = 0.869$ – 0.953), which became more pronounced at higher precipitation (median $\Delta = 0.124$ – 0.134 ; $\Pr(\Delta > 0) = 0.779$ – 0.790). In *P. autumnalis*, growth was context-dependent: unfenced plots shifted from negative to positive effects across the precipitation gradient (median $\Delta = -0.275$ to 0.144 ; $\Pr(\Delta > 0) = 0.169$ – 0.775), while fencing generally amplified benefits at low to moderate precipitation (median $\Delta = 0.020$ – 0.390 ; $\Pr(\Delta > 0) = 0.559$ – 0.927), but became negative at higher precipitation (median $\Delta = -0.262$ to -0.322 ; $\Pr(\Delta > 0) = 0.056$ – 0.072).

Epichloë endophytes consistently increased inflorescence production across the precipitation gradient (Figs. 5). By contrast, spikelet production was negatively affected by *Epichloë* under low precipitation but increased under wetter conditions, with effects varying among species and depending on herbivore exclusion (S-4; Figs. S-5). Inflorescence benefits in *A. hyemalis* and *E. virginicus* were minimal under dry conditions but increased with precipitation, with only minor influence from herbivore exclusion. In *P. autumnalis*, strong inflorescence benefits occurred under fencing at moderate-to-high precipitation (median $\Delta = 0.47$ – 1.54 ; $\Pr(\Delta > 0) = 0.96$ – 1.00 ; Table S-5). Spikelet production showed a similar pattern: *E. virginicus* and *P. autumnalis* experienced weak or negative effects under low precipitation (median $\Delta = 0.10$ – 0.40 ; $\Pr(\Delta > 0) = 0.23$ – 0.40), shifting to positive effects at higher precipitation, with fencing amplifying these benefits across the gradient (median $\Delta = 0.95$ – 1.56 ; $\Pr(\Delta > 0) = 0.79$ – 0.99 ; Table S-6).

Abiotic factors as the primary drivers of fitness components across the stress gradient

Grass-endophyte symbiosis, whether mutualistic or parasitic, was primarily modulated by climatic variation rather than by herbivore pressure. Across all vital rates (survival, growth, inflorescence, and spikelet production), models comparing different drivers of the endophyte–host interaction consistently identified the *Endophyte* $\times$ *Climate* model as the best predictor of host demography, based on expected log predictive density (ELPD; higher values indicate better fit; Table 1). For instance, for survival, the *Endophyte* $\times$ *Climate* model outperformed the *Endophyte* $\times$ *Herbivory* model ($\Delta\text{ELPD} = -2.1$, $\text{SE} = 2.8$), and including the three-way *Endophyte* $\times$ *Climate* $\times$ *Herbivory* interaction further decreased predictive performance ($\Delta\text{ELPD} =$

—5.0, SE = 1.9). Similar patterns were observed for growth, inflorescence, and spikelet production, indicating that climatic context is the primary driver of variation in the demographic consequences of endophyte symbiosis, whereas herbivory has weaker or context-dependent effects.

Discussion

Drawing on three years of field experiments that crossed endophyte infection (infected vs. free) with herbivory exclusion, and using Bayesian statistical models, we evaluated how biotic and abiotic stressors jointly shape the demographic effects of vertically transmitted *Epichloë* endophytes in three cool-season perennial grasses across a natural climatic gradient in the United States. Our results indicate that grass–endophyte symbioses generally enhance host performance under favorable (e.g. wetter) conditions, whereas their effects can be neutral or even detrimental under high abiotic stress, consistent with the inhibition scenario. Across species and fitness components, climatic variation emerged as the dominant driver of endophyte-mediated effects, while herbivory had a secondary or context-dependent influence. These results underscore that the outcome of grass–*Epichloë* symbiosis shifts along abiotic stress gradients (Bronstein, 1994; Chamberlain et al., 2014; Saikkonen et al., 1998), providing empirical support for environmentally contingent mutualism in a broadly distributed host–symbiont system.

Endophyte presence often enhances host performance under favorable or moderately stressful conditions, a pattern widely documented, a pattern widely documented across grass–*Epichloë* systems (Clay, 1990). Endophytes can increase host growth, reproduction, and resistance to herbivores (Rodriguez et al., 2009), but these benefits are most apparent when the host has sufficient resources, such as water and nutrients, to support the fungus (Gibert et al., 2012). Mechanistically, the host must allocate carbon and nutrients to maintain the endophyte, so under resource-rich conditions, the costs of symbiosis are outweighed by the benefits, resulting in a net positive effect. In our field experiment, we observed higher growth and survival of endophyte-infected grasses in wetter sites, consistent with previously documented positive effects of endophytes under favorable conditions. Other potential consequences of endophyte symbiosis, including enhanced nutrient acquisition (Malinowski et al., 2000), have yet to be explored in this system. Current work is assessing host fitness responses to endophytes, a critical step toward understanding how abiotic and biotic stressors interact to shape these symbiotic relationships.

While endophyte infection enhanced host performance under favorable or moderately stressful conditions, its effects diminished and ultimately became costly under high abiotic stress, even in fenced plots where herbivores were excluded. This pattern indicates that drought-driven resource limitation can impose costs on the symbiosis independent of biotic

stress (Bronstein, 2001; Johnson et al., 1997). This contrasts with other studies showing that context-dependent benefits of endophytes are often driven by herbivore pressure. For example, in *Bromus setifolia*, high herbivore pressure favors the symbiosis because the benefits outweigh the costs, whereas low herbivore pressure or other high costs select against endophyte infection (Rodriguez et al., 2009). Our results suggest that similar context dependence can arise from abiotic factors, such as drought, independent of herbivory. Under low-precipitation conditions, reduced water availability constrains nutrient uptake and increases the relative metabolic burden of maintaining the endophyte, potentially tipping the cost–benefit balance against endophyte infected plants (Ahlholm et al., 2002; Bronstein, 2001; Faeth, 2002). In addition, severe environmental stress can activate plant defense pathways—such as elevated phytohormone production and reactive oxygen species (ROS) that may inhibit endophyte growth or reduce its functional contributions to the host (Bastías and Gundel, 2023; Eaton et al., 2011). Consequently, even in the absence of herbivores, endophyte-mediated benefits may erode or become negative in resource-limited environments, consistent with the reduced performance of infected plants at the driest sites in our study. Although we found no evidence of infection-related costs for inflorescences across the precipitation gradient, we did detect a small but consistent cost for spikelet production at the driest sites, suggesting that different reproductive components may vary in sensitivity to drought-mediated shifts in host–endophyte interactions (Cheplick and Faeth, 2009). These abiotic costs also appear to affect reproduction: while inflorescence production showed no detectable cost across the precipitation gradient, spikelet production exhibited a small but consistent cost under the driest conditions, suggesting that resource limitations may disproportionately affect certain reproductive components in infected plants.

Across species and fitness components, climatic variation emerged as the dominant driver of endophyte-mediated effects, whereas herbivory played a secondary or context-dependent role. Precipitation strongly shaped host demography in our system, with drier sites constraining growth and survival. Contrary to the predictions of the stress gradient hypothesis (SGH), which suggests that biotic interactions should become more positive under stressful conditions, endophyte-mediated benefits were reduced under high abiotic stress (Louthan et al., 2015). Although most empirical tests of the SGH have focused on plant–plant interactions, recent syntheses show that support for the SGH is actually strongest among microbial interactions (Hammarlund and Harcombe, 2019; Piccardi et al., 2019), with bacterial systems showing greater conformity to SGH predictions than any other kingdom (Adams et al., 2022). This contrast highlights that plant–microbe mutualisms, including fungal endophytes, may not follow the same stress-dependency patterns documented in plant community ecology, and that endophyte benefits can decline rather than intensify under severe environmental stress.

³⁴⁷ **Acknowledgements**

³⁴⁸ This research was supported by National Science Foundation Division of Environmental
³⁴⁹ Biology awards.

References

- Adams, A. E., Besozzi, E. M., Shahrokhi, G., and Patten, M. A. (2022). A case for associational resistance: Apparent support for the stress gradient hypothesis varies with study system. *Ecology Letters*, 25(1):202–217.
- Afkhami, M. E. and Rudgers, J. A. (2008). Symbiosis lost: imperfect vertical transmission of fungal endophytes in grasses. *The American Naturalist*, 172(3):405–416.
- Ahlholm, J. U., Helander, M., Lehtimäki, S., Wäli, P., and Saikkonen, K. (2002). Vertically transmitted fungal endophytes: different responses of host-parasite systems to environmental conditions. *Oikos*, 99(1):173–183.
- Amijee, F., Tinker, P., and Stribley, D. (1989). The development of endomycorrhizal root systems: Vii. a detailed study of effects of soil phosphorus on colonization. *New Phytologist*, 111(3):435–446.
- Atala, C., Acuña-Rodríguez, I. S., Torres-Díaz, C., and Molina-Montenegro, M. A. (2022). Fungal endophytes improve the performance of host plants but do not eliminate the growth/defence trade-off. *New Phytologist*, 235(2).
- Bastías, D. A. and Gundel, P. E. (2023). Plant stress responses compromise mutualisms with epichloë endophytes. *Journal of Experimental Botany*, 74(1):19–23.
- Bastias, D. A., Martínez-Ghersa, M. A., Ballaré, C. L., and Gundel, P. E. (2017). Epichloë fungal endophytes and plant defenses: not just alkaloids. *Trends in Plant Science*, 22(11):939–948.
- Bronstein, J. L. (1994). Conditional outcomes in mutualistic interactions. *Trends in ecology & evolution*, 9(6):214–217.
- Bronstein, J. L. (2001). The exploitation of mutualisms. *Ecology letters*, 4(3):277–287.
- Chamberlain, S. A., Bronstein, J. L., and Rudgers, J. A. (2014). How context dependent are species interactions? *Ecology letters*, 17(7):881–890.
- Cheplick, G. P. and Faeth, S. H. (2009). *Ecology and evolution of the grass-endophyte symbiosis*. OUP USA.
- Clay, K. (1990). Fungal endophytes of grasses. *Annual review of Ecology and Systematics*, pages 275–297.

- Clay, K., Holah, J., and Rudgers, J. A. (2005). Herbivores cause a rapid increase in hereditary symbiosis and alter plant community composition. *Proceedings of the National Academy of Sciences*, 102(35):12465–12470.
- Clay, K. and Schardl, C. (2002). Evolutionary origins and ecological consequences of endophyte symbiosis with grasses. *the american naturalist*, 160(S4):S99–S127.
- Eaton, C. J., Cox, M. P., and Scott, B. (2011). What triggers grass endophytes to switch from mutualism to pathogenism? *Plant Science*, 180(2):190–195.
- Emery, S. M., Thompson, D., and Rudgers, J. A. (2010). Variation in endophyte symbiosis, herbivory and drought tolerance of amrophila breviligulata populations in the great lakes region. *The American Midland Naturalist*, 163(1):186–196.
- Faeth, S. H. (2002). Are endophytic fungi defensive plant mutualists? *Oikos*, 98(1):25–36.
- Fowler, J. C., Donald, M. L., Bronstein, J. L., and Miller, T. E. (2023). The geographic footprint of mutualism: How mutualists influence species’ range limits. *Ecological Monographs*, 93(1):e1558.
- Fowler, J. C., Moutouama, J., and Miller, T. E. (2025). Increasing prevalence of plant-fungal symbiosis across two centuries of environmental change. *Global Change Biology*, 31(11):e70577.
- GBIF (2025a). Occurrence download for *Agrostis hyemalis*. Accessed 5 November 2025.
- GBIF (2025b). Occurrence download for *Elymus virginicus*. Accessed 5 November 2025.
- GBIF (2025c). Occurrence download for *Poa autumnalis*. Accessed 5 November 2025.
- Giauque, H., Connor, E. W., and Hawkes, C. V. (2019). Endophyte traits relevant to stress tolerance, resource use and habitat of origin predict effects on host plants. *New Phytologist*, 221(4):2239–2249.
- Gibert, A., Volaire, F., Barre, P., and Hazard, L. (2012). A fungal endophyte reinforces population adaptive differentiation in its host grass species. *New Phytologist*, 194(2):561–571.
- Gundel, P. E., Sun, P., Charlton, N. D., Young, C. A., Miller, T. E., and Rudgers, J. A. (2020). Simulated folivory increases vertical transmission of fungal endophytes that deter herbivores and alter tolerance to herbivory in poa autumnalis. *Annals of Botany*, 125(6):981–991.
- Hammarlund, S. P. and Harcombe, W. R. (2019). Refining the stress gradient hypothesis in a microbial community. *Proceedings of the National Academy of Sciences*, 116(32):15760–15762.

407 Hart, E. M. and Bell, K. (2015). *prism: Download data from the Oregon prism project*. R package
408 version 0.0.6.

409 Hoeksema, J. D. and Bruna, E. M. (2015). Context-dependent outcomes of mutualistic
410 interactions. *Mutualism*, 10:181–202.

411 Johnson, N. C., Graham, J. H., and Smith, F. A. (1997). Functioning of mycorrhizal associations
412 along the mutualism–parasitism continuum. *The New Phytologist*, 135(4):575–585.

413 Klironomos, J. N. (2003). Variation in plant response to native and exotic arbuscular
414 mycorrhizal fungi. *Ecology*, 84(9):2292–2301.

415 Koh, S., Vicari, M., Ball, J., Rakocevic, T., Zaheer, S., Hik, D., and Bazely, D. (2006). Rapid
416 detection of fungal endophytes in grasses for large-scale studies. *Functional Ecology*, pages
417 736–742.

418 Leuchtmann, A., Bacon, C. W., Schardl, C. L., White Jr, J. F., and Tadych, M. (2014). Nomenclat-
419 ural realignment of neotyphodium species with genus epichloë. *Mycologia*, 106(2):202–215.

420 Liu, M. (2023). Fungi canadenses no. 351: Epichloe elymi. *Canadian Journal of Plant Pathology*,
421 45(3):300–303.

422 Louthan, A. M., Doak, D. F., and Angert, A. L. (2015). Where and when do species interactions
423 set range limits? *Trends in ecology & evolution*, 30(12):780–792.

424 Malinowski, D. P., Alloush, G. A., and Belesky, D. P. (2000). Leaf endophyte neotyphodium
425 coenophialum modifies mineral uptake in tall fescue. *Plant and Soil*, 227(1):115–126.

426 Oberhofer, M., Güsewell, S., and Leuchtmann, A. (2014). Effects of natural hybrid and
427 non-hybrid epichloë endophytes on the response of hordelymus europaeus to drought
428 stress. *New Phytologist*, 201(1):242–253.

429 Piccardi, P., Vessman, B., and Mitri, S. (2019). Toxicity drives facilitation between 4 bacterial
430 species. *Proceedings of the National Academy of Sciences*, 116(32):15979–15984.

431 Porter, S. S., Bantay, R., Friel, C. A., Garoutte, A., Gdanetz, K., Ibarreta, K., Moore, B. M.,
432 Shetty, P., Siler, E., and Friesen, M. L. (2020). Beneficial microbes ameliorate abiotic and
433 biotic sources of stress on plants. *Functional Ecology*, 34(10):2075–2086.

434 R Core Team (2023). *R: A Language and Environment for Statistical Computing*. R Foundation
435 for Statistical Computing, Vienna, Austria.

- Rodriguez, R., White Jr, J., Arnold, A. E., and Redman, a. R. a. (2009). Fungal endophytes: diversity and functional roles. *New phytologist*, 182(2):314–330.
- Rudgers, J. A., Fletcher, R. A., Olivas, E., Young, C. A., Charlton, N. D., Pearson, D. E., and Maron, J. L. (2016). Long-term ungulate exclusion reduces fungal symbiont prevalence in native grasslands. *Oecologia*, 181(4):1151–1161.
- Ruiz-lazona, J. M., Azcón, R., and Palma, J. (1996). Superoxide dismutase activity in arbuscular mycorrhizal lactuca sativa plants subjected to drought stress. *New Phytologist*, 134(2):327–333.
- Saikkonen, K., Faeth, S. H., Helander, M., and Sullivan, T. (1998). Fungal endophytes: a continuum of interactions with host plants. *Annual review of Ecology and Systematics*, 29(1):319–343.
- Saikkonen, K., Saari, S., and Helander, M. (2010). Defensive mutualism between plants and endophytic fungi? *Fungal Diversity*, 41(1):101–113.
- Saikkonen, K., Wäli, P., Helander, M., and Faeth, S. H. (2004). Evolution of endophyte–plant symbioses. *Trends in plant science*, 9(6):275–280.
- Schardl, C. L., Leuchtmann, A., and Spiering, M. J. (2004). Symbioses of grasses with seedborne fungal endophytes. *Annu. Rev. Plant Biol.*, 55(1):315–340.
- Shaw, R. B. (2011). *Guide to Texas grasses*. Texas A&M University Press.
- Silva, L. A. and Zanella, G. (2024). Robust leave-one-out cross-validation for high-dimensional bayesian models. *Journal of the American Statistical Association*, 119(547):2369–2381.
- Soudzilovskaia, N. A., van der Heijden, M. G., Cornelissen, J. H., Makarov, M. I., Onipchenko, V. G., Maslov, M. N., Akhmetzhanova, A. A., and van Bodegom, P. M. (2015). Quantitative assessment of the differential impacts of arbuscular and ectomycorrhiza on soil carbon cycling. *New Phytologist*, 208(1):280–293.
- Stan Development Team (2024). RStan: the R interface to Stan. R package version 2.32.6.
- Thirkell, T. J., Pastok, D., and Field, K. J. (2020). Carbon for nutrient exchange between arbuscular mycorrhizal fungi and wheat varies according to cultivar and changes in atmospheric carbon dioxide concentration. *Global change biology*, 26(3):1725–1738.
- Vega, F. E. (2008). Insect pathology and fungal endophytes. *Journal of invertebrate pathology*, 98(3):277–279.

- 464 Vehtari, A., Gelman, A., and Gabry, J. (2017). Practical bayesian model evaluation using
465 leave-one-out cross-validation and waic. *Statistics and computing*, 27:1413–1432.
- 466 White Jr, J. F. (1994). Endophyte-host associations in grasses. xx. structural and reproductive
467 studies of *epichloë amarillans* sp. nov. and comparisons to *e. typhina*. *Mycologia*,
468 86(4):571–580.
- 469 Yule, K. M., Miller, T. E., and Rudgers, J. A. (2013). Costs, benefits, and loss of vertically
470 transmitted symbionts affect host population dynamics. *Oikos*, 122(10):1512–1520.

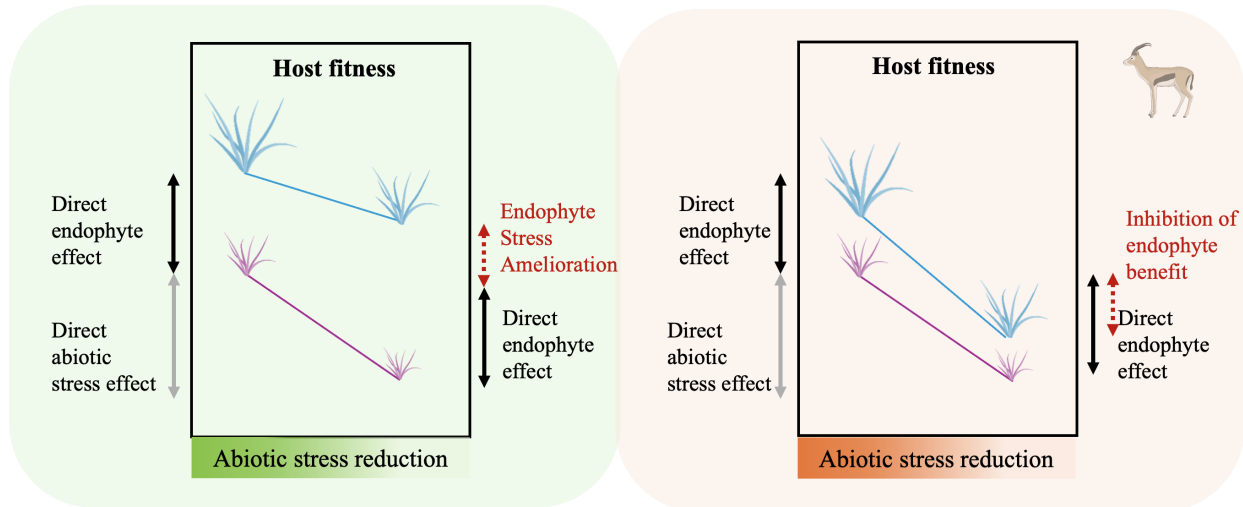


Figure 1: Conceptual framework illustrating the predicted effects of abiotic stress, biotic stress, and endophyte-mediated mutualism on host plant fitness (adapted from (Porter et al., 2020)). (a) **Stress-amelioration scenario:** Endophytes are predicted to enhance host performance under increasing abiotic stress (e.g., low precipitation) by mitigating drought-related physiological costs. Abiotic stress is the primary driver of host fitness, and endophyte-mediated benefits become more pronounced as stress intensifies. (b) **Inhibition scenario:** When abiotic stress is combined with an additional biotic stressor herbivory the positive direct effect of endophytes is expected to weaken. Under high combined stress (abiotic + biotic), herbivory suppresses or interferes with drought-mitigation benefits, reducing or eliminating the symbiotic advantage and potentially resulting in a net cost of harboring an endophyte. Colors are used solely for visualization: blue lines represent endophyte-infected plants and violet lines represent endophyte-free plants. Endophyte infection cannot be visually detected in living plants.

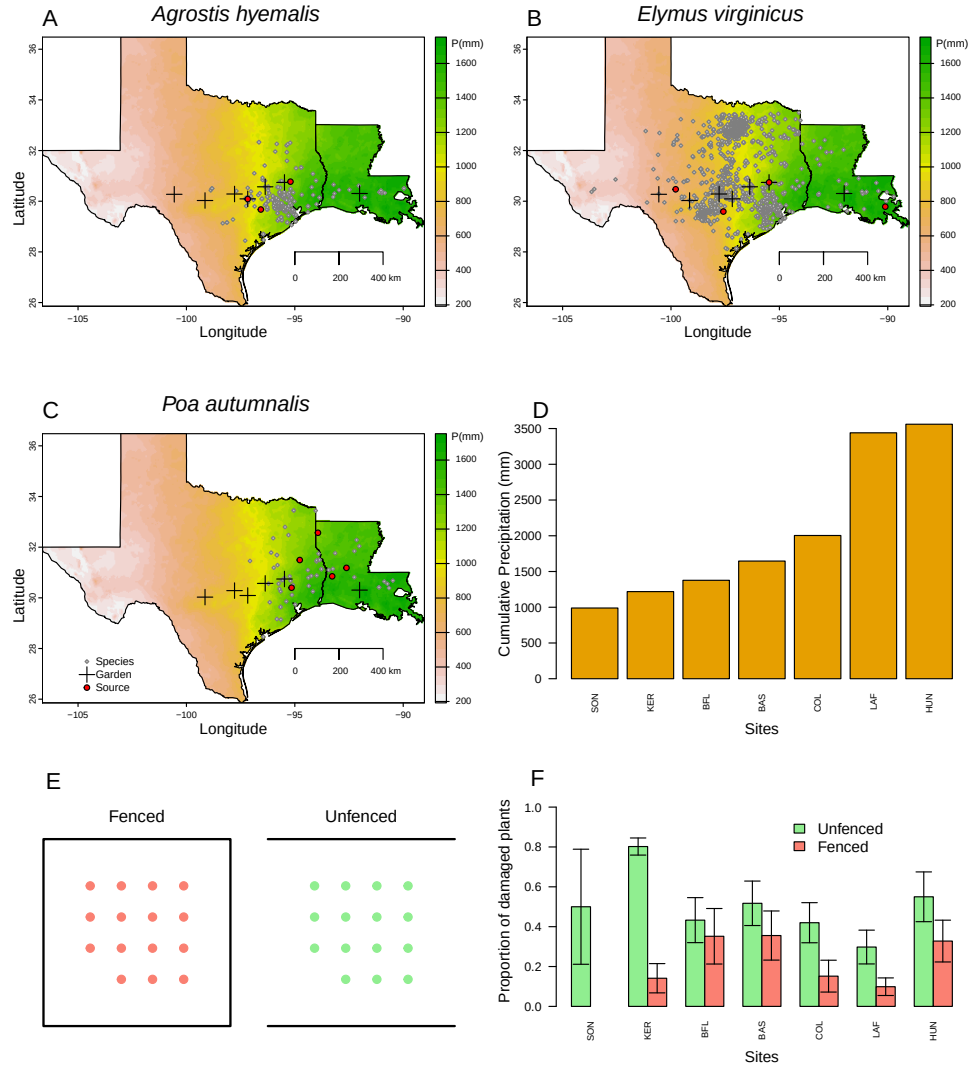


Figure 2: Distribution of common garden sites along a natural aridity gradient from Texas to Louisiana, US for (A) *Agrostis hyemalis*, (B) *Elymus virginicus*, and (C) *Poa autumnalis*. Climate data in A–C are based on 30-year normals of climate predictions (1993–2023) from PRISM data (Hart and Bell, 2015), which illustrate the precipitation gradient across sites (scale on the right is in mm). Red dots indicate the locations of source populations, while grey dots represent Global Biodiversity Information Facility (GBIF) occurrence records for each species within the study area (GBIF, 2025a,b,c). D) Cumulative precipitation during the study period (May 2023 – June 2025) at each site, ordered west to east. Values are also derived from PRISM climate data (Hart and Bell, 2015).

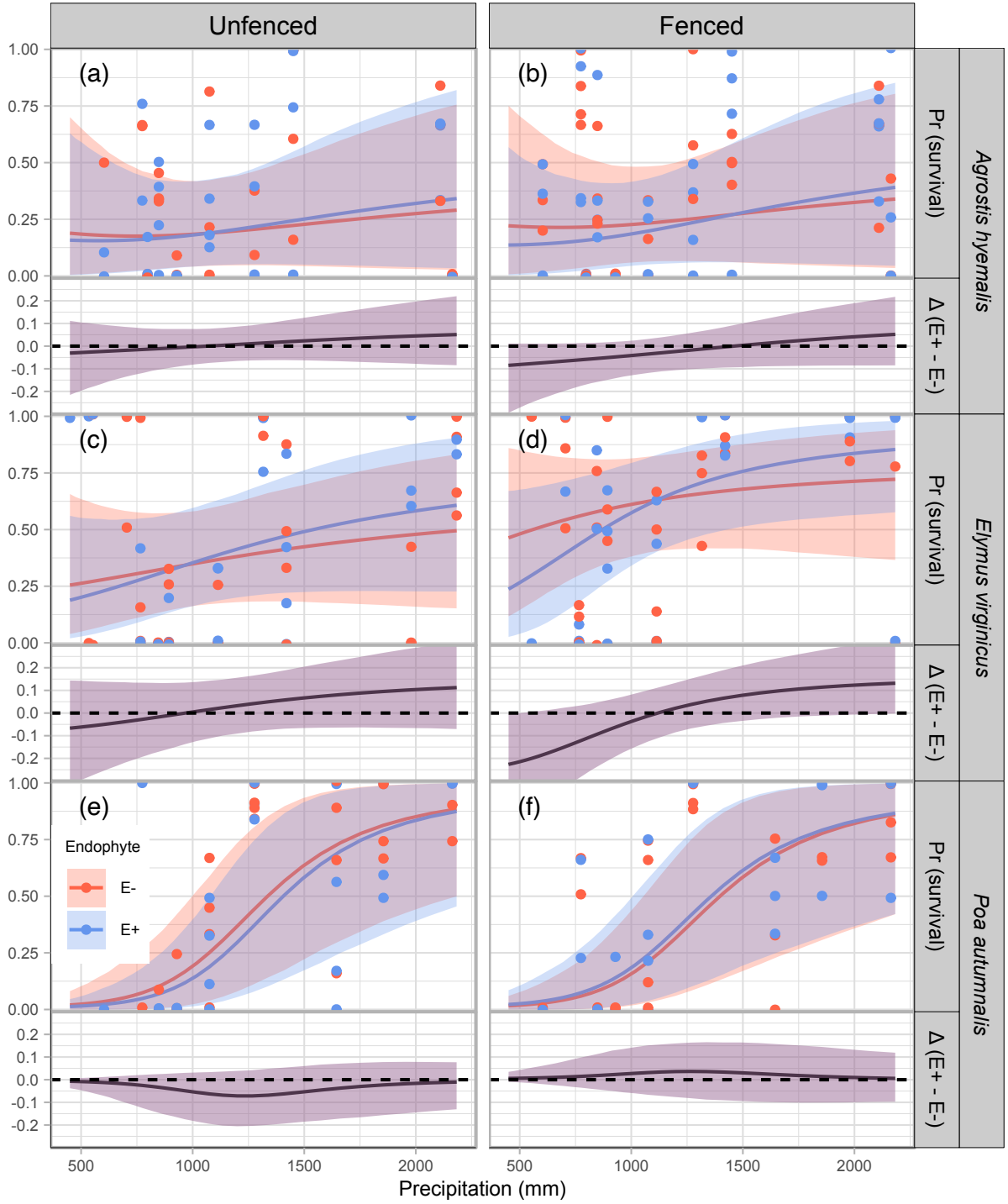


Figure 3: Predicted survival rate (top panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed survival per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

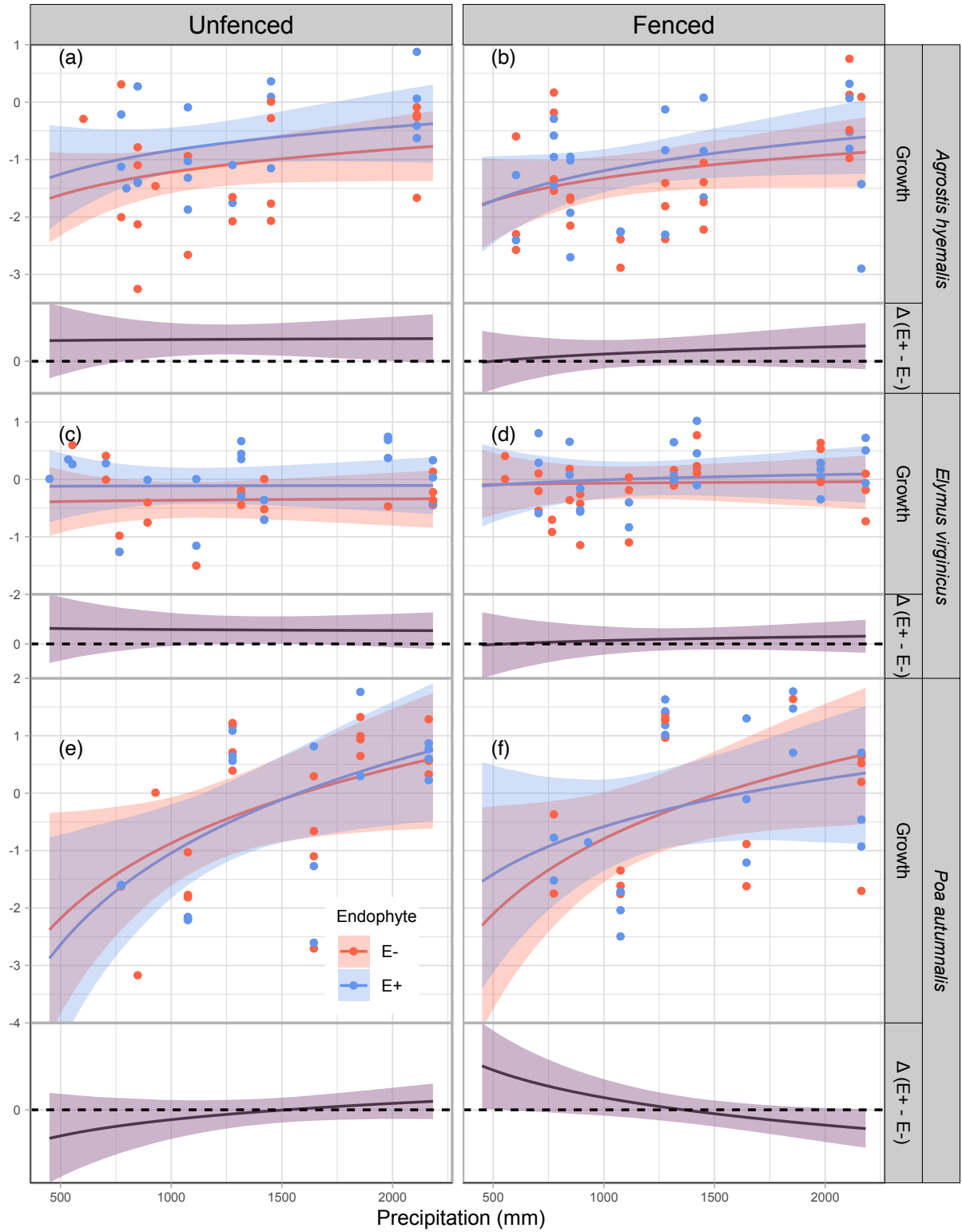


Figure 4: Predicted relative growth from year to the next year (top panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed growth per plot, jittered for visualization. The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

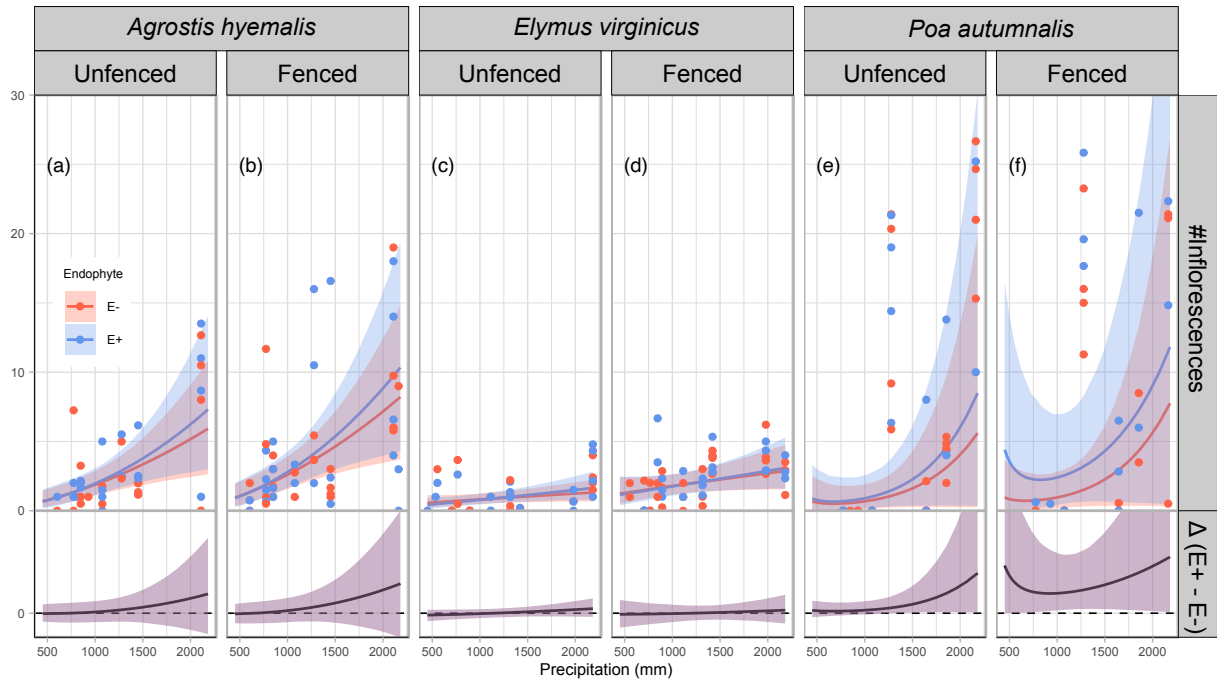


Figure 5: Predicted number of inflorescences (upper panels) and the difference between endophyte-infected (E^+) and endophyte-free (E^-) plants ($\Delta(E^+ - E^-)$, lower panels) are shown across a gradient of precipitation. Shaded areas represent 90% credible intervals. Points indicate observed means, jittered for visualization. Panels are grouped by species and herbivory treatment (unfenced = with herbivory; fenced = herbivory excluded).

Table 1: Model comparison of drivers of grass- endophyte symbiosis across four fitness components (vital rates). Δ ELPD indicates the difference in expected log predictive density relative to the best-fitting model; SE is the standard error of the difference.

Vital rate	Model type	Δ ELPD	SE
Survival	Endophyte $\times$ Climate	0.00	0.00
	Endophyte $\times$ Herbivory	-2.11	2.81
	Endophyte $\times$ Climate $\times$ Herbivory	-4.98	1.89
Growth	Endophyte $\times$ Climate	0.00	0.00
	Endophyte $\times$ Herbivory	-0.19	2.81
	Endophyte $\times$ Climate $\times$ Herbivory	-1.81	2.75
Inflorescence	Endophyte $\times$ Climate	0.00	0.00
	Endophyte $\times$ Herbivory	-2.62	3.53
	Endophyte $\times$ Climate $\times$ Herbivory	-3.78	2.63
Spikelet	Endophyte $\times$ Climate	0.00	0.00
	Endophyte $\times$ Herbivory	-2.01	3.82
	Endophyte $\times$ Climate $\times$ Herbivory	-3.16	2.43

472 **S.1 Supporting Information**

473 **List of Supplementary Figures**

474	S1	Abiotic conditions across the common garden experiments	35
475	S2	Posterior predictive checks for each vital rate	36

⁴⁷⁶ **List of Supplementary Tables**

⁴⁷⁷ S1 Geographic coordinates and descriptions of source populations for study species 28

478 S.1.1 Supporting Methods

479 S.1.2 Candidate models for abiotic and biotic drivers of cool-season grass 480 species

- 481 1. **Endophyte × Climate model:** includes endophyte status (endo), precipitation (P), and
482 their interaction:

$$483 \mu_i = \beta_0[\text{Sp}_i] + \beta_{\text{endo}}[\text{Sp}_i]\text{endo}_i + \beta_P[\text{Sp}_i]P_i + \beta_{\text{endo} \times P}[\text{Sp}_i]\text{endo}_iP_i \\ 484 + \phi_{\text{site}[i], \text{Sp}_i} + \omega_{\text{source}[i], \text{Sp}_i} + \rho_{\text{plot}[i], \text{Sp}_i}$$

- 485 2. **Endophyte × Herbivory model:** includes endophyte status (endo), herbivory (herb), and
486 their interaction:

$$487 \mu_i = \beta_0[\text{Sp}_i] + \beta_{\text{endo}}[\text{Sp}_i]\text{endo}_i + \beta_{\text{herb}}[\text{Sp}_i]\text{herb}_i + \beta_{\text{endo} \times \text{herb}}[\text{Sp}_i]\text{endo}_i\text{herb}_i \\ 488 + \phi_{\text{site}[i], \text{Sp}_i} + \omega_{\text{source}[i], \text{Sp}_i} + \rho_{\text{plot}[i], \text{Sp}_i}$$

- 489 3. **Endophyte × Climate × Herbivory:** combines all predictors and interactions up to the
490 three-way interaction:

$$491 \mu_i = \beta_0[\text{Sp}_i] + \beta_P[\text{Sp}_i]P_i + \beta_{\text{endo}}[\text{Sp}_i]\text{endo}_i + \beta_{\text{herb}}[\text{Sp}_i]\text{herb}_i \\ 492 + \beta_{\text{endo} \times P}[\text{Sp}_i]\text{endo}_iP_i + \beta_{\text{herb} \times P}[\text{Sp}_i]\text{herb}_iP_i + \beta_{\text{herb} \times \text{endo}}[\text{Sp}_i]\text{herb}_i\text{endo}_i \\ 493 + \beta_{\text{endo} \times \text{herb} \times P}[\text{Sp}_i]\text{endo}_i\text{herb}_iP_i + \phi_{\text{site}[i], \text{Sp}_i} + \omega_{\text{source}[i], \text{Sp}_i} + \rho_{\text{plot}[i], \text{Sp}_i}$$

494 In these models, the β coefficients represent species-specific effects. The intercept $\beta_0[\text{Sp}_i]$
495 corresponds to the baseline value of the vital rate for species Sp_i in the absence of predictors.
496 $\beta_{\text{endo}}[\text{Sp}_i]$ and $\beta_{\text{herb}}[\text{Sp}_i]$ describe the effects of endophyte presence and herbivory, respectively.
497 The linear effect of precipitation is captured by $\beta_P[\text{Sp}_i]$, while interactions are represented
498 by $\beta_{\text{endo} \times P}[\text{Sp}_i]$, $\beta_{\text{herb} \times P}[\text{Sp}_i]$, $\beta_{\text{endo} \times \text{herb}}[\text{Sp}_i]$, and the three-way interaction $\beta_{\text{endo} \times \text{herb} \times P}[\text{Sp}_i]$.
499 All models include hierarchical random effects for site-year, source population, and plot,
500 with species-specific random effects $\phi_{\text{site}[i], \text{Sp}_i} \sim \mathcal{N}(0, \sigma_{\text{site}})$, $\omega_{\text{source}[i], \text{Sp}_i} \sim \mathcal{N}(0, \sigma_{\text{source}})$, and
501 $\rho_{\text{plot}[i], \text{Sp}_i} \sim \mathcal{N}(0, \sigma_{\text{plot}})$.

S.1.3 Supporting Tables

Supptable S1: Geographic coordinates and brief descriptions of source populations for *Agrostis hyemalis* (AGHY), *Elymus virginicus* (ELVI), and *Poa autumnalis* (POAU).

Species	Population	Latitude	Longitude	Description
AGHY	SJC	30.771969	-95.200620	Ecolab property
AGHY	COHR	29.672833	-96.567517	Columbus, TX herbarium record
AGHY	LPEF	30.085383	-97.172508	Lost Pines
POAU	LA7	30.849660	-93.276772	DeRidder, LA
POAU	LA1	32.568125	-93.932976	Jean Lafitte National Historical Park and Preserve
POAU	LA2	31.183040	-92.616969	Jean Lafitte National Historical Park and Preserve
POAU	SFAEF	31.492158	-94.772200	Stephen F. Austin Experimental Forest, TX
POAU	WB	30.399346	-95.150026	Sam Houston National Forest
ELVI	HUNT	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	SHS	30.741797	-95.474508	Sam Houston State University Field Station
ELVI	RL5	30.471717	-99.783803	Texas Tech University, Junction
ELVI	PALM	29.591623	-97.584781	Palmetto State Park, TX
ELVI	JLP	29.785105	-90.114933	Jean Lafitte Park, outside of New Orleans

Table S-1: Common garden and field experiment sites used for reciprocal transplant and environmental manipulation studies.

Site	Code	Latitude	Longitude
University of Louisiana Ecology Center	LAF	30.304773	-92.012401
Center for Biological Field Studies	HUN	30.742815	-95.476881
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
1335 Medina Hwy, Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
TAMU Sonora Station	SON	30.273467	-100.561463

Table S-2: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOOCV). ELPD difference is relative to the best model.

Vital rate	Model type	ELPD difference	SE difference
Survival	Quadratic	0	0.000
Survival	Linear	-1.230	0.947
Growth	Quadratic	0	0.000
Growth	Linear	-0.544	0.924
Flowering	Quadratic	0	0.000
Flowering	Linear	-0.763	0.956
Spikelet	Quadratic	0	0.000
Spikelet	Linear	-0.097	0.957

Table S-3: Posterior summaries of $\Delta (E^+ - E^-)$ for survival at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
<i>Agrostis hyemalis</i>	Unfenced	765.76	-0.01055	-0.12081	0.07964	0.3723
<i>Agrostis hyemalis</i>	Unfenced	848.28	-0.00814	-0.10368	0.07597	0.3980
<i>Agrostis hyemalis</i>	Unfenced	1113.25	0.00173	-0.07149	0.07904	0.5239
<i>Agrostis hyemalis</i>	Unfenced	1644.36	0.02230	-0.06632	0.14693	0.6981
<i>Agrostis hyemalis</i>	Unfenced	2163.28	0.03508	-0.08410	0.21801	0.7237
<i>Agrostis hyemalis</i>	Fenced	765.76	-0.04897	-0.17775	0.01302	0.0961
<i>Agrostis hyemalis</i>	Fenced	848.28	-0.04477	-0.15803	0.01516	0.1076
<i>Agrostis hyemalis</i>	Fenced	1113.25	-0.02642	-0.11372	0.03630	0.2266
<i>Agrostis hyemalis</i>	Fenced	1644.36	0.01141	-0.08847	0.12851	0.5950
<i>Agrostis hyemalis</i>	Fenced	2163.28	0.03789	-0.08484	0.21422	0.7222
<i>Elymus virginicus</i>	Unfenced	765.76	-0.02574	-0.19555	0.13388	0.3799
<i>Elymus virginicus</i>	Unfenced	848.28	-0.01565	-0.17000	0.13230	0.4250
<i>Elymus virginicus</i>	Unfenced	1113.25	0.01937	-0.10368	0.14589	0.6070
<i>Elymus virginicus</i>	Unfenced	1644.36	0.07749	-0.06409	0.22970	0.8217
<i>Elymus virginicus</i>	Unfenced	2163.28	0.10622	-0.07003	0.31076	0.8445
<i>Elymus virginicus</i>	Fenced	765.76	-0.12411	-0.28674	0.03254	0.0992
<i>Elymus virginicus</i>	Fenced	848.28	-0.09198	-0.23593	0.04709	0.1418
<i>Elymus virginicus</i>	Fenced	1113.25	-0.00179	-0.11213	0.11041	0.4882
<i>Elymus virginicus</i>	Fenced	1644.36	0.08855	-0.02165	0.24020	0.9069
<i>Elymus virginicus</i>	Fenced	2163.28	0.11327	-0.00261	0.32611	0.9456
<i>Poa autumnalis</i>	Unfenced	765.76	-0.00979	-0.12238	0.02097	0.1853
<i>Poa autumnalis</i>	Unfenced	848.28	-0.01778	-0.14683	0.02450	0.1735
<i>Poa autumnalis</i>	Unfenced	1113.25	-0.05636	-0.20042	0.03503	0.1429
<i>Poa autumnalis</i>	Unfenced	1644.36	-0.02891	-0.17290	0.07018	0.2804
<i>Poa autumnalis</i>	Unfenced	2163.28	-0.00174	-0.13201	0.07681	0.4319
<i>Poa autumnalis</i>	Fenced	765.76	0.00399	-0.03466	0.09583	0.6685
<i>Poa autumnalis</i>	Fenced	848.28	0.00739	-0.04282	0.11317	0.6803
<i>Poa autumnalis</i>	Fenced	1113.25	0.02557	-0.06906	0.15496	0.7063
<i>Poa autumnalis</i>	Fenced	1644.36	0.01678	-0.09915	0.15764	0.6297
<i>Poa autumnalis</i>	Fenced	2163.28	0.00181	-0.09727	0.12122	0.5663

Table S-4: Posterior summaries of $\Delta (E^+ - E^-)$ for growth across precipitation quantiles for each species and herbivore treatment. Positive median values indicate beneficial effects of endophytes; negative medians indicate costly effects.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
<i>Agrostis hyemalis</i>	Unfenced	773.91	0.37006	-0.00924	0.75437	0.9456
<i>Agrostis hyemalis</i>	Unfenced	1075.09	0.37940	0.10078	0.65111	0.9866
<i>Agrostis hyemalis</i>	Unfenced	1316.25	0.38541	0.11668	0.64408	0.9915
<i>Agrostis hyemalis</i>	Unfenced	1979.35	0.39167	0.01393	0.77022	0.9539
<i>Agrostis hyemalis</i>	Unfenced	2163.28	0.39436	-0.02748	0.81310	0.9400
<i>Agrostis hyemalis</i>	Fenced	773.91	0.07955	-0.23563	0.39683	0.6612
<i>Agrostis hyemalis</i>	Fenced	1075.09	0.13909	-0.10277	0.37262	0.8279
<i>Agrostis hyemalis</i>	Fenced	1316.25	0.17455	-0.07319	0.41710	0.8781
<i>Agrostis hyemalis</i>	Fenced	1979.35	0.24676	-0.11528	0.60922	0.8733
<i>Agrostis hyemalis</i>	Fenced	2163.28	0.26153	-0.13527	0.66118	0.8657
<i>Elymus virginicus</i>	Unfenced	773.91	0.25869	-0.11938	0.62862	0.8693
<i>Elymus virginicus</i>	Unfenced	1075.09	0.24977	-0.02146	0.51978	0.9343
<i>Elymus virginicus</i>	Unfenced	1316.25	0.24421	0.00389	0.48413	0.9530
<i>Elymus virginicus</i>	Unfenced	1979.35	0.23379	-0.05653	0.52131	0.9096
<i>Elymus virginicus</i>	Unfenced	2163.28	0.23175	-0.08415	0.54666	0.8877
<i>Elymus virginicus</i>	Fenced	773.91	0.03164	-0.30420	0.37363	0.5644
<i>Elymus virginicus</i>	Fenced	1075.09	0.06458	-0.16044	0.29000	0.6825
<i>Elymus virginicus</i>	Fenced	1316.25	0.08512	-0.10777	0.27375	0.7682
<i>Elymus virginicus</i>	Fenced	1979.35	0.12472	-0.12987	0.37637	0.7903
<i>Elymus virginicus</i>	Fenced	2163.28	0.13352	-0.15046	0.41561	0.7788
<i>Poa autumnalis</i>	Unfenced	773.91	-0.27505	-0.74845	0.20422	0.1688
<i>Poa autumnalis</i>	Unfenced	1075.09	-0.14041	-0.44570	0.16567	0.2237
<i>Poa autumnalis</i>	Unfenced	1316.25	-0.05910	-0.29128	0.17786	0.3413
<i>Poa autumnalis</i>	Unfenced	1979.35	0.10761	-0.15864	0.37505	0.7388
<i>Poa autumnalis</i>	Unfenced	2163.28	0.14420	-0.16050	0.44856	0.7749
<i>Poa autumnalis</i>	Fenced	773.91	0.39049	-0.05133	0.83526	0.9270
<i>Poa autumnalis</i>	Fenced	1075.09	0.16142	-0.12313	0.45470	0.8212
<i>Poa autumnalis</i>	Fenced	1316.25	0.02003	-0.20891	0.25959	0.5587
<i>Poa autumnalis</i>	Fenced	1979.35	-0.26222	-0.54759	0.03589	0.0723
<i>Poa autumnalis</i>	Fenced	2163.28	-0.32245	-0.64896	0.01200	0.0558

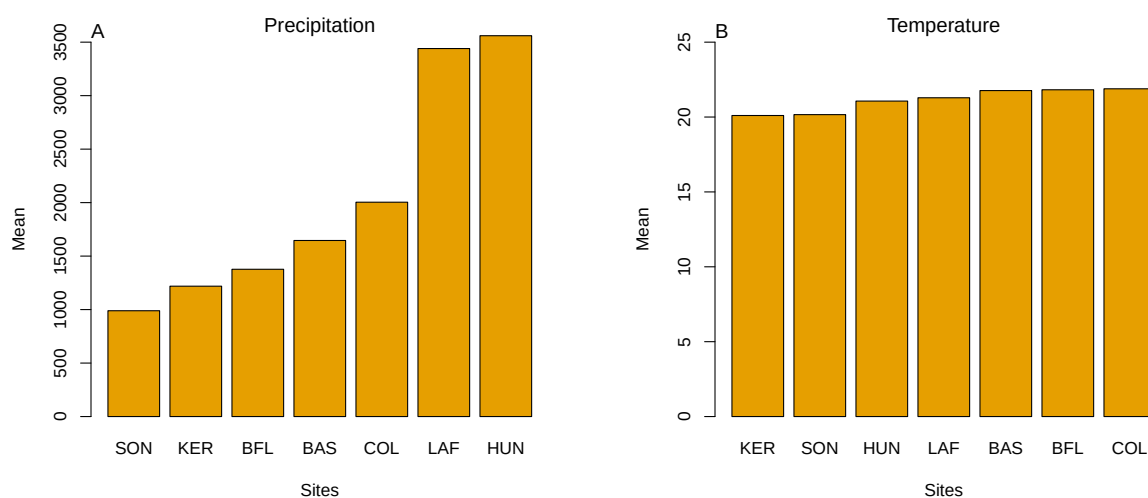
Table S-5: Posterior summaries of Δ ($E^+ - E^-$) for flowering output at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
<i>Agrostis hyemalis</i>	Unfenced	773.91	-0.6824	-0.6904	0.7822	0.4889
<i>Agrostis hyemalis</i>	Unfenced	1075.09	0.0286	-0.6483	0.9444	0.6018
<i>Agrostis hyemalis</i>	Unfenced	1316.25	0.4663	-0.6218	1.2633	0.6909
<i>Agrostis hyemalis</i>	Unfenced	1979.35	1.3488	-1.2041	4.0378	0.7552
<i>Agrostis hyemalis</i>	Unfenced	2163.28	1.5409	-1.4924	5.3973	0.7522
<i>Agrostis hyemalis</i>	Fenced	773.91	-0.6824	-0.7351	0.9077	0.5238
<i>Agrostis hyemalis</i>	Fenced	1075.09	0.0286	-0.6302	1.2286	0.6696
<i>Agrostis hyemalis</i>	Fenced	1316.25	0.4663	-0.6108	1.7763	0.7609
<i>Agrostis hyemalis</i>	Fenced	1979.35	1.3488	-1.3208	5.6835	0.7965
<i>Agrostis hyemalis</i>	Fenced	2163.28	1.5409	-1.7235	7.3637	0.7922
<i>Elymus virginicus</i>	Unfenced	773.91	-0.6824	-0.4386	0.2723	0.3289
<i>Elymus virginicus</i>	Unfenced	1075.09	0.0286	-0.3390	0.3120	0.4611
<i>Elymus virginicus</i>	Unfenced	1316.25	0.4663	-0.2792	0.3903	0.5915
<i>Elymus virginicus</i>	Unfenced	1979.35	1.3488	-0.2491	0.8798	0.7902
<i>Elymus virginicus</i>	Unfenced	2163.28	1.5409	-0.2647	1.0691	0.8082
<i>Elymus virginicus</i>	Fenced	773.91	-0.6824	-0.8003	0.7325	0.4283
<i>Elymus virginicus</i>	Fenced	1075.09	0.0286	-0.6057	0.5997	0.4688
<i>Elymus virginicus</i>	Fenced	1316.25	0.4663	-0.4972	0.5628	0.5208
<i>Elymus virginicus</i>	Fenced	1979.35	1.3488	-0.6334	1.0660	0.6272
<i>Elymus virginicus</i>	Fenced	2163.28	1.5409	-0.7473	1.3341	0.6318
<i>Poa autumnalis</i>	Unfenced	773.91	-0.6824	-0.1587	0.7919	0.7062
<i>Poa autumnalis</i>	Unfenced	1075.09	0.0286	-0.0711	0.9392	0.8638
<i>Poa autumnalis</i>	Unfenced	1316.25	0.4663	0.0137	1.3447	0.9629
<i>Poa autumnalis</i>	Unfenced	1979.35	1.3488	0.0793	6.8120	0.9926
<i>Poa autumnalis</i>	Unfenced	2163.28	1.5409	0.0645	10.6951	0.9858
<i>Poa autumnalis</i>	Fenced	773.91	-0.6824	0.0907	5.6618	0.9981
<i>Poa autumnalis</i>	Fenced	1075.09	0.0286	0.2492	4.3096	0.9998
<i>Poa autumnalis</i>	Fenced	1316.25	0.4663	0.3164	4.7790	0.9999
<i>Poa autumnalis</i>	Fenced	1979.35	1.3488	0.1612	11.8419	0.9976
<i>Poa autumnalis</i>	Fenced	2163.28	1.5409	0.0888	15.3801	0.9848

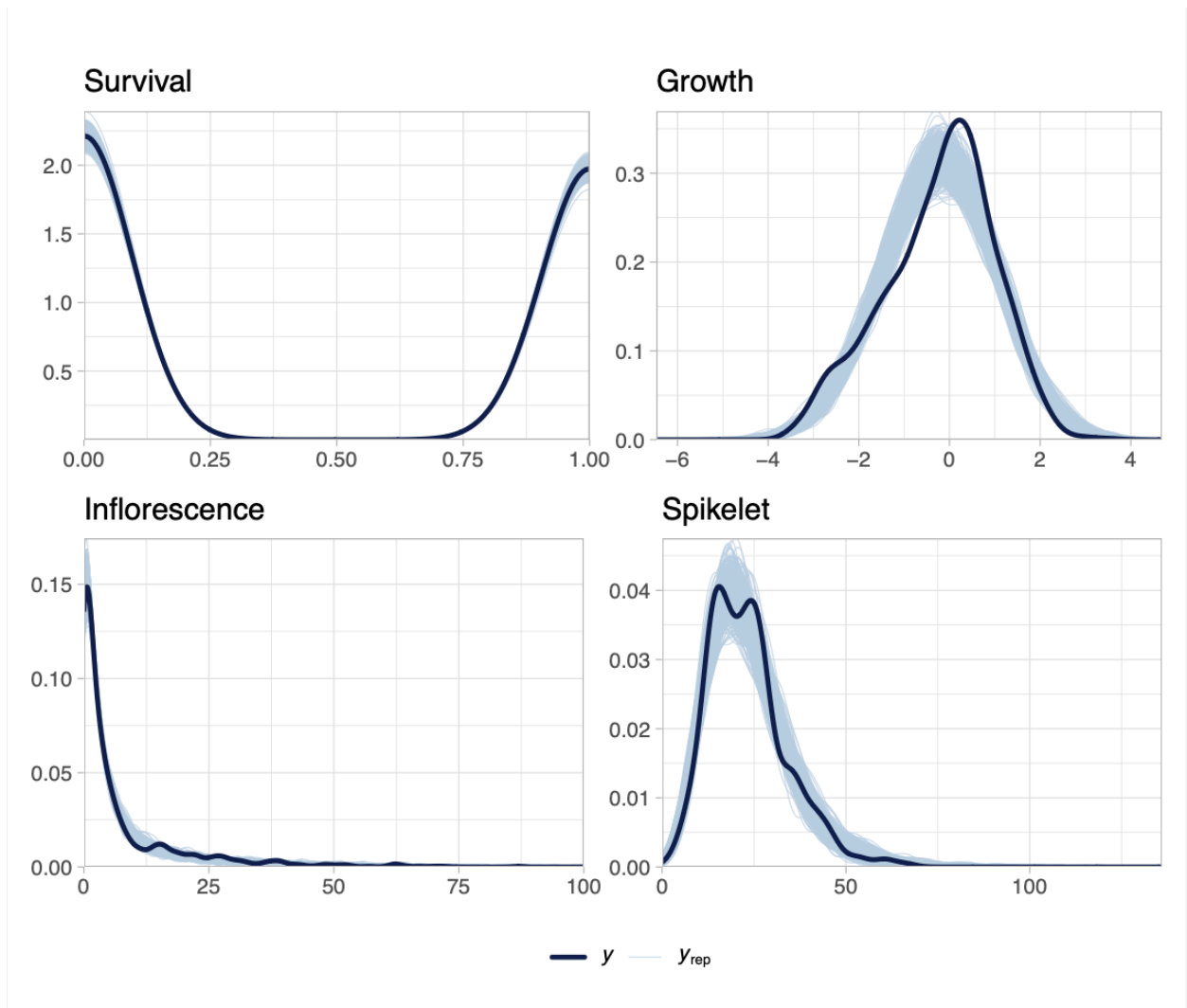
Table S-6: Posterior summaries of Δ ($E^+ - E^-$) at precipitation quantiles for each species and herbivore exclusion treatment. Median, 90% credible interval, and posterior probability that $\Delta > 0$ are shown. Herbivore exclusion: Fenced = herbivores excluded, Unfenced = herbivores had access.

Species	Herbivore exclusion	Precipitation (mm)	Median Δ	Lower 90%	Upper 90%	$P(\Delta > 0)$
<i>Elymus virginicus</i>	Unfenced	1113.25	0.1040	-4.4921	2.4788	0.3213
<i>Elymus virginicus</i>	Unfenced	1276.99	0.4008	-3.6975	2.7993	0.4012
<i>Elymus virginicus</i>	Unfenced	1644.36	0.9477	-2.5664	3.8884	0.6089
<i>Elymus virginicus</i>	Unfenced	2163.28	1.5409	-2.0652	6.1715	0.7733
<i>Elymus virginicus</i>	Unfenced	2182.71	1.5603	-2.0610	6.2851	0.7760
<i>Elymus virginicus</i>	Fenced	1113.25	0.1040	-4.2244	1.7020	0.2288
<i>Elymus virginicus</i>	Fenced	1276.99	0.4008	-3.0122	2.0977	0.3714
<i>Elymus virginicus</i>	Fenced	1644.36	0.9477	-1.2591	3.8036	0.7853
<i>Elymus virginicus</i>	Fenced	2163.28	1.5409	-0.4058	7.9946	0.9282
<i>Elymus virginicus</i>	Fenced	2182.71	1.5603	-0.3886	8.1667	0.9288
<i>Poa autumnalis</i>	Unfenced	1113.25	0.1040	-3.2874	1.7757	0.3134
<i>Poa autumnalis</i>	Unfenced	1276.99	0.4008	-2.7023	1.8008	0.3654
<i>Poa autumnalis</i>	Unfenced	1644.36	0.9477	-1.7415	2.3862	0.5832
<i>Poa autumnalis</i>	Unfenced	2163.28	1.5409	-1.9577	5.4441	0.7572
<i>Poa autumnalis</i>	Unfenced	2182.71	1.5603	-1.9811	5.5917	0.7592
<i>Poa autumnalis</i>	Fenced	1113.25	0.1040	0.3203	6.9989	0.9668
<i>Poa autumnalis</i>	Fenced	1276.99	0.4008	0.6863	6.6041	0.9828
<i>Poa autumnalis</i>	Fenced	1644.36	0.9477	0.8105	6.7216	0.9873
<i>Poa autumnalis</i>	Fenced	2163.28	1.5409	-0.9449	8.6686	0.8957
<i>Poa autumnalis</i>	Fenced	2182.71	1.5603	-1.0369	8.7690	0.8906

S.1.4 Supporting Figures



Suppfigure S1: Abiotic conditions across the seven study sites. (A) Mean annual precipitation showing the pronounced east-west aridity gradient from mesic Louisiana to arid Texas. (B) Mean annual temperature across sites, which does not differ much. Panels highlight that precipitation, rather than temperature, constitutes the primary abiotic gradient in this system.



Suppfigure S2: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y), and light blue lines show replicated datasets generated from posterior predictions (y_{rep}). The strong overlap across survival, growth, flowering, and spikelet production indicates adequate model fit for all vital rates.

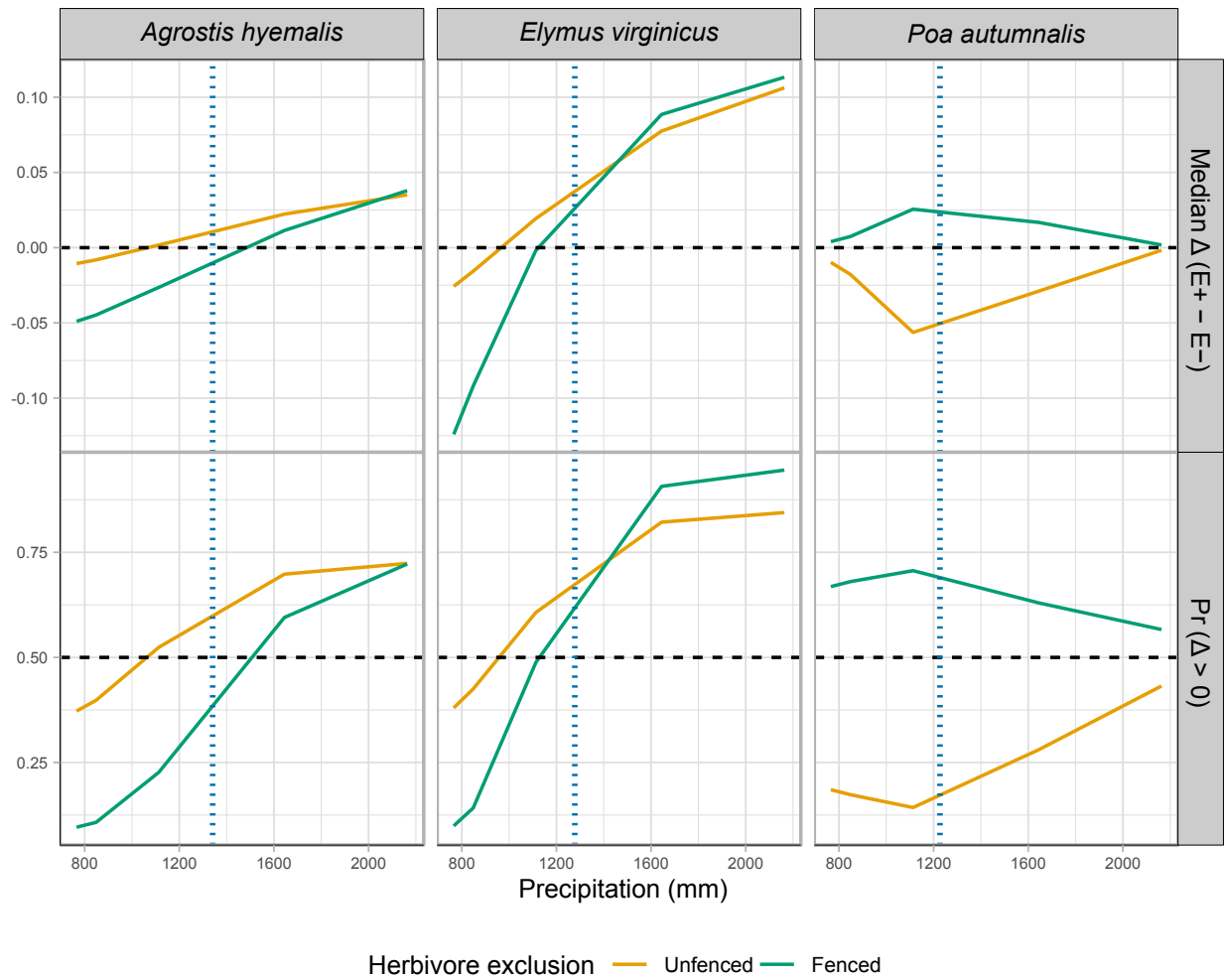


Figure S-1: Predicted effects of endophyte presence (E^+) versus absence (E^-) on survival across precipitation gradients for three cool-season grasses. Solid lines show the median $\Delta (E^+ - E^-)$, or the posterior probability that $\Delta > 0$ under fenced and unfenced conditions. Horizontal dashed lines indicate no effect ($\Delta = 0$) or a 50% probability of benefit. Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

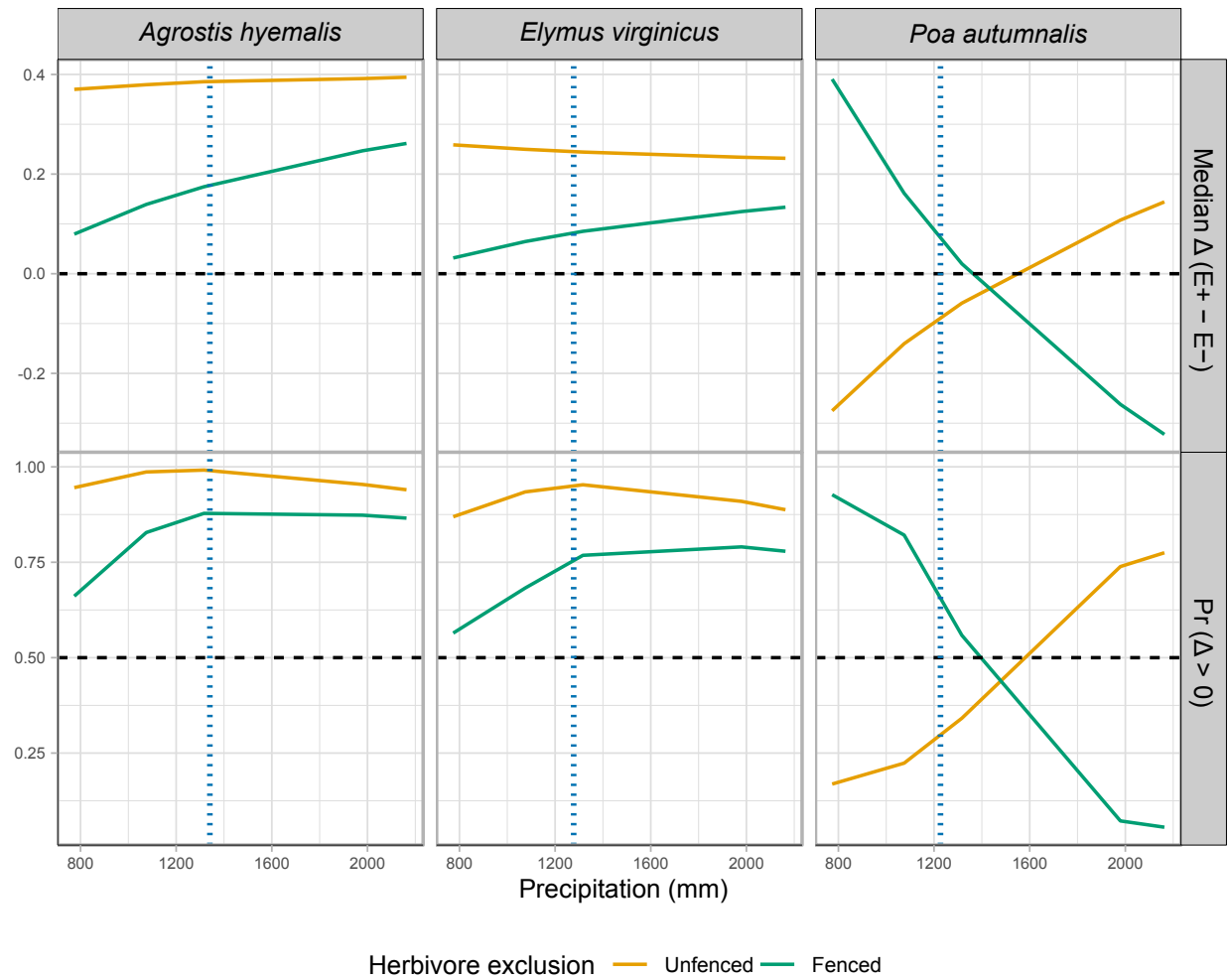


Figure S-2: Effects of precipitation and endophyte presence on growth across three cool-season grass species. Lines show the predicted difference in growth between endophyte-present (E^+) and endophyte-absent (E^-) plants (Median Δ , top row) and the posterior probability that this difference is positive ($\Pr(\Delta > 0)$, bottom row) under fenced and unfenced treatments. Predictions were calculated at the 10th, 25th, 50th, 75th, and 90th percentiles of precipitation observed in the study. Horizontal dashed lines indicate $\Delta = 0$ (top row) and $\Pr(\Delta > 0) = 0.5$ (bottom row). Vertical dashed lines represent the precipitation values at the most westerly natural occurrence of each species, based on GBIF records.

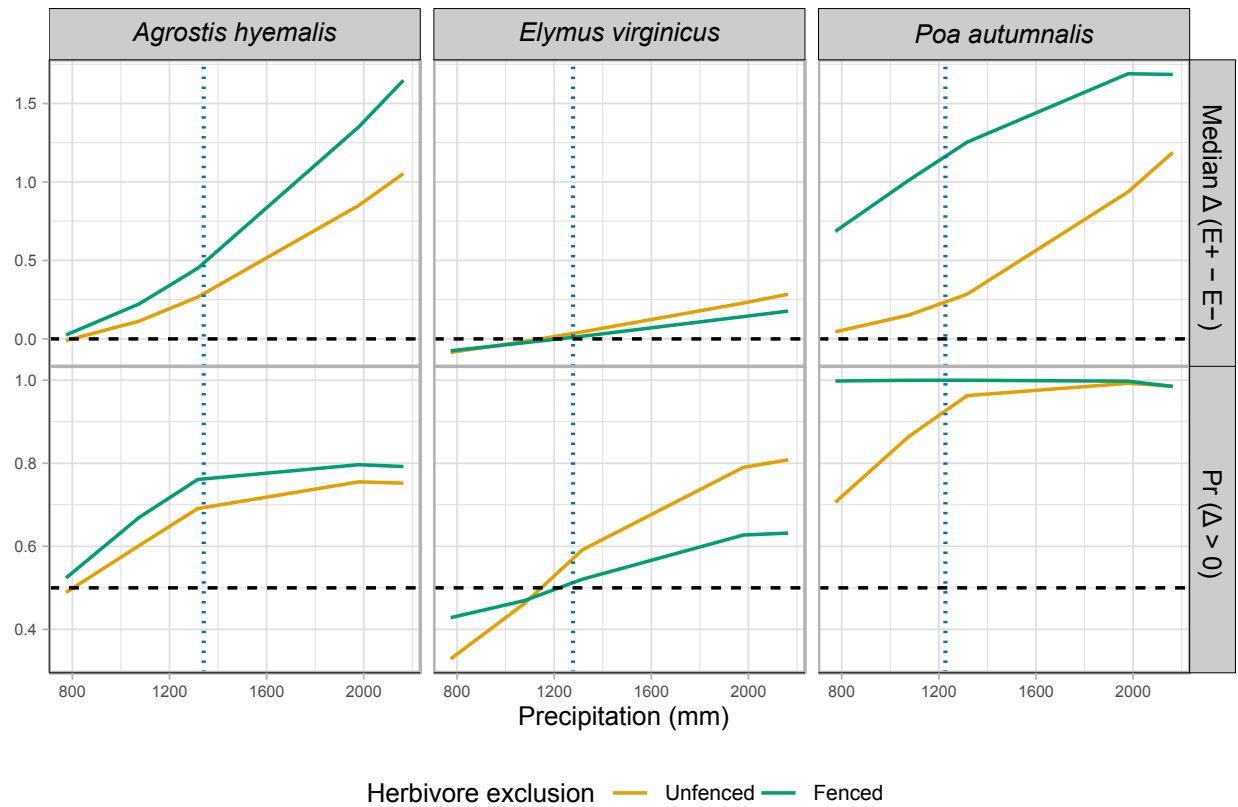


Figure S-3: Effects of endophyte infection on flowering output across a precipitation gradient for three cool-season grass species. Lines show the median posterior difference $\Delta(E^+ - E^-)$ and the posterior probability that $\Delta > 0$ under fenced (herbivores excluded) and unfenced (herbivores present) conditions. Horizontal dashed lines indicate $\Delta = 0$ for median differences and $\Pr(\Delta > 0) = 0.5$. Endophyte effects were generally positive for *Agrostis hyemalis* and *Poa autumnalis*, but for *Elymus virginicus* benefits were context-dependent, with costs at lower precipitation

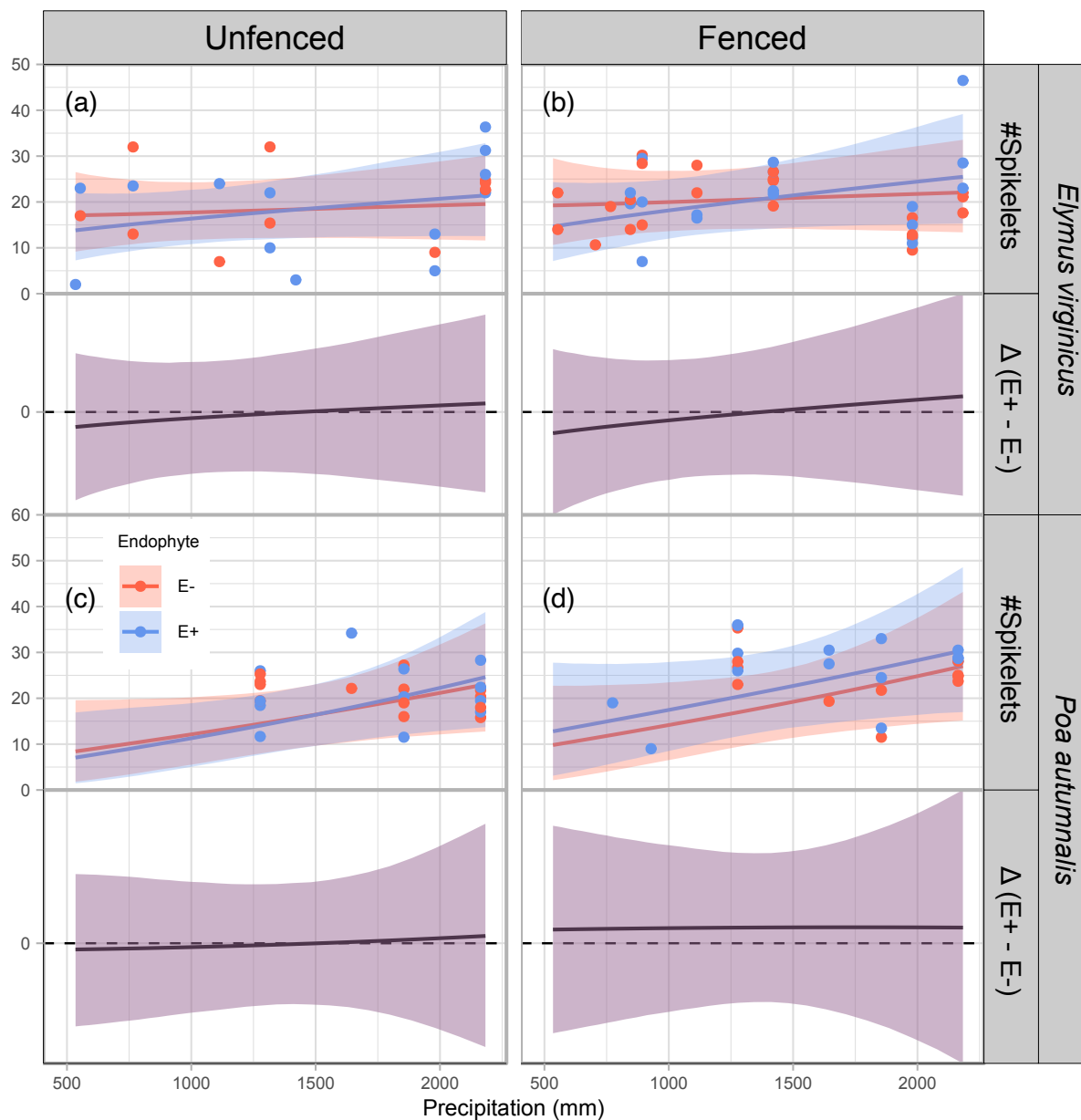


Figure S-4: Predicted number of spikelets (upper panels) and endophyte effect ($\Delta = E^+ - E^-$, bottom panels) across precipitation gradients. Lines show posterior means for E^+ (blue) and E^- (red), with shading indicating 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for only two species (see Methods for details). The dashed line at $\Delta = 0$ indicates no endophyte effect. Panels are organized by species and herbivory treatment (Fenced, Unfenced).

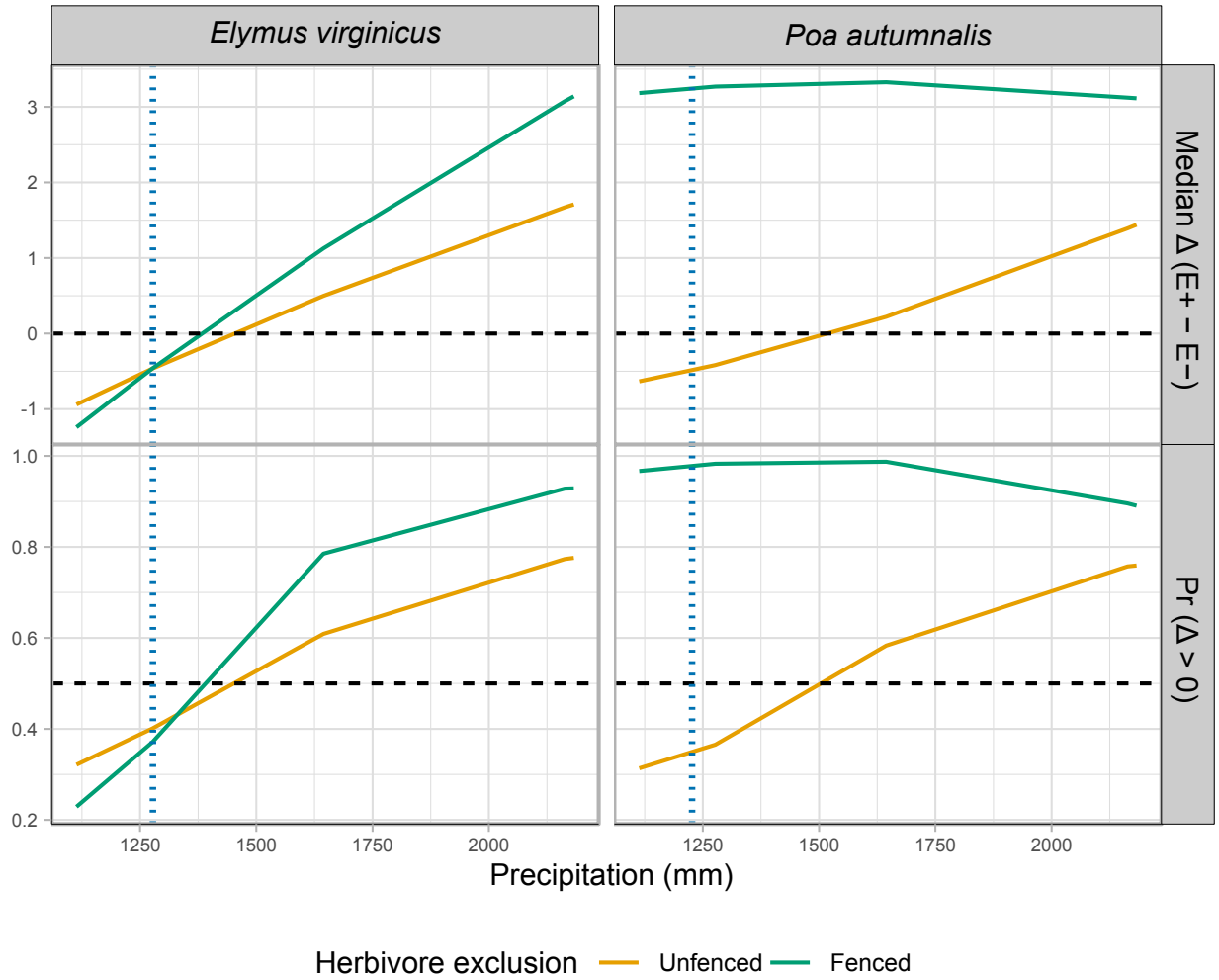


Figure S-5: Effects of precipitation and endophyte presence on spike (reproductive output) for three cool-season grass species. Median $\Delta (E^+ - E^-)$ and posterior probability $\text{Pr}(\Delta > 0)$ are shown across precipitation quantiles. Herbivore exclusion treatments are indicated by color (Fenced = herbivores excluded, Unfenced = herbivores had access). Horizontal dashed lines indicate $\Delta = 0$ for medians and $\text{Pr}(\Delta > 0) = 0.5$ for probabilities. Facets separate metrics and species.